Optimizing Premium Pricing: Predictive Analytics for Insurance Loss

DSO 530: Applied Modern Statistical Learning Methods

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Business Problem 📌

Challenge: Inaccurate premiums cause adverse selection - losing good customers, keeping risky ones.
Goal: Accurately predict claim losses to ensure fair pricing and a profitable portfolio.

- **Data Overview** 📊
- **39,928 training records (28 variables)**
- **13,310 test records (23 variables)**

Target Variables construction:

	LC	HALC	CS
What	Cost per claim	Historically adjusted LC	Did they Claim
Range	0 – 35,191	0 – 78,131	0 or 1
Mean	$64.56	$117.26	11.23% (1)

- **Data Cleaning** 🧹

453 corrupt rows dropped from train (X.23 = 0, X.26 = 0, X.27 = NaN), **140 missing X.27 imputed** in test via Random Forest Classifier – 130 Petrol, 10 Diesel

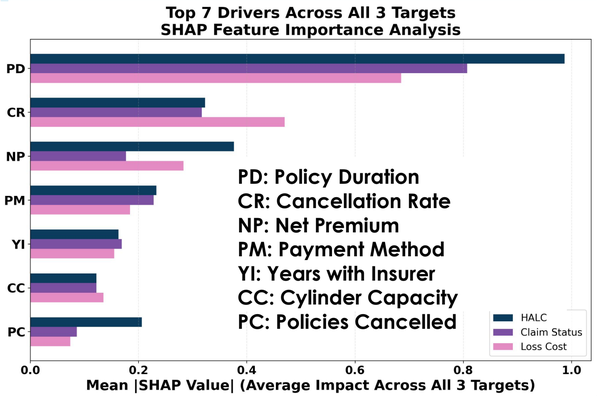
Predicting LC & HALC 🤖

Model Approach	LC Val MSE	HALC Val MSE	Status
Tweedie GLM	324,175	1,026,099	Baseline
XGBoost Tweedie	317,720	1,009,443	Better
LightGBM Tweedie	317,852	1,008,616	Better
Regularized LightGBM	316,612	1,007,285	Best single
Stacking Ensemble	316,465	1,005,405	WINNER

Predicting Claim Status 🤖

Model Approach	Train AUC	Val AUC	vs. Baseline
Logistic Regression	0.7152	0.6965	Baseline
Random Forest	0.8046	0.7521	+7.98%.
XGBoost Classifier	0.8664	0.7884	+12.78%.

Model Interpretation – SHAP Analysis 💭



Key Findings: Policy Duration & Cancellation Rate dominate all 3 Targets; behavioral features outperform vehicle specs, challenging conventional insurance pricing.

Business Insights & Recommendations 💡

Policy Duration: The Loyalty Effect

- **Finding:** Longer policies = lower risk (22.4% → 3.3%).
- **Action:** Reward loyalty - long-term customers are ~7× less likely to claim.

Cancellation Rates: The Sweet Spot

- **Finding:** U-shaped risk - lowest at 0–20% cancellations (~5.5–6.1%).
- **Action:** Target 0–20% group; penalize >50% cancellations.

Driver Age: Risk Peaks

- **Finding:** Highest risk at 18–25; smaller spike at 46–55.
- **Action:** Higher premiums for <35; analyze mid-career spike.

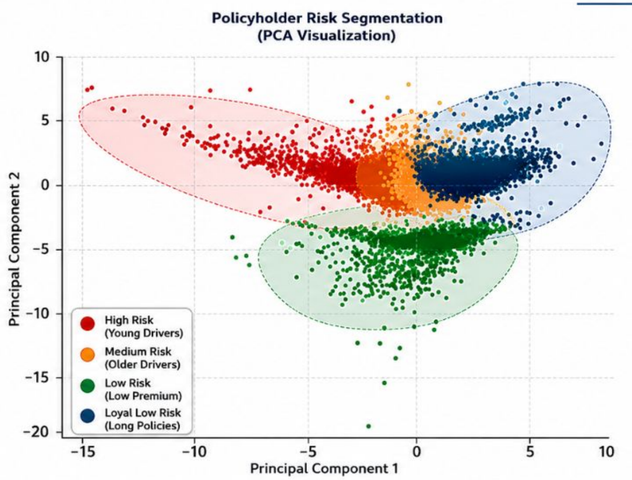
Vehicle Type: Risk Differences

- **Finding:** Vans highest risk; agriculture 0%, motorbikes low (4.1%).
- **Action:** Raise van premiums; competitively price low-risk vehicles.

Premium Adequacy: HALC Warning

- **Finding:** HALC loss ratios nearing risk levels in $0–200 & $400–600 bands.
- **Action:** Increase premiums 10–15% in those segments.

Innovation: Customer Risk Segmentation 🚗



Segment (Clusters)	Profile (yr)	Count	Avg LC	Claim Rate	Avg Premium
C0	Young Drivers (35)	13,881	$89.4	14.3%	$361
C1	Older Drivers (54)	17,548	$64.1	11.7%	$326
C2	Low Premium (45)	2,557	$23.1	4.2%	$146
C3	Loyal Long Term (52)	5,489	$22.5	5.1%	$301

🔴 High Risk 🟠 Medium Risk 🟢 Low Risk